

Problem Set 1

ILA 386 · Data Science for Linguists

Data Wrangling and Cleaning

For this exercise, you will work with two main data files: `data_stress.csv` and `data_proficiency.csv`. These files represent what data might look like when you download it from an experiment platform such as Gorilla, survey software such as Qualtrics, and so on. The `data_stress.csv` in particular contains many columns that you likely will not need, so the first thing you need to do is make sense of the data and decide what columns to keep and discard and what rows to keep and discard.

The data come from a study we ran in the Speech Learning Lab where we tested participants' ability to identify lexical stress differences such as *canto* ('I sing') and *cantó* ('He/She sang'). On this task, Spanish learners heard a verb and were asked to associate the verb with the correct subject pronoun. For instance, if they heard *canto*, they would select 'yo' because that is the subject pronoun that would be paired with that verb based on its ending (its morphology). Stimuli were produced by many different Spanish speakers and contained target items (like the *canto* and *cantó* distinction previously mentioned) and distractor items to check that participants understood how to conjugate verbs in Spanish. We needed to check this because we wanted to be sure that errors on our target items were due to challenges with lexical stress perception rather than lack of knowledge of Spanish verb morphology.

The `data_stress.csv` has a response column indicating what response the participant gave but it does not (yet) have a column indicating if participants responded correctly. You need to generate that. The answers are stored in `data_answers.csv`, which you need to combine with `data_stress.csv` using `merge()`. You will need to figure out how to align the data sets to enable a correct merge, so that you end up with an 'answer' column in your stress data frame. You can then use an `ifelse()` expression with 'answer' and 'response' to generate a new 'score' variable, where you code responses as correct (1) or incorrect (0). Once you get that, you should create a mean score per participant on distractor and target items.

You need to do the same for the `data_proficiency.csv` file. That is, you need to decide which columns and rows to keep and which to discard, create a score variable using the 'response' and 'answer' columns, and then create a mean score for each participant. Once you've got the two data frames, you are ready to do one final merge into a combined dataset that will be ready for analysis.

Checklist of Wrangling and Cleaning Decisions

Real experimental data are rarely tidy. As you work, you will need to deal with each of the issues below. Some have a single right answer that you need to find; others are judgment calls, where what matters is that you make a reasonable choice and justify it. Address every item in your code and explain what you did and why in a short comment.

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Stress data (`data_stress.csv` and `data_answers.csv`)

- **Rows.** Each trial takes up several rows (instructions, audio events, stimulus and response screens, and so on). Which rows contain the participant's actual response?
- **Columns.** Which columns do you need for merging, scoring, and grouping? Which can you safely drop?
- **Merge key.** Which column(s) will you use to link each trial to its answer? Check that your key uniquely identifies (indexes) each item in `data_answers.csv`, and that its values are written exactly the same way in both files (look closely at spelling, capitalization, accents, and character encoding). Some information appears in more than one column, so explore your options before committing to one.
- **Response and answer coding.** Before you use `ifelse()`, compare the `unique()` values of the 'response' and 'answer' columns. Would a correct response actually match its answer?
- **Missing responses.** Some trials timed out without a response. Will you count these as incorrect or exclude them? Check how `ifelse()` and `mean()` handle NA values, so that the choice you make is the one your code actually carries out.

Proficiency data (`data_proficiency.csv`)

- **Rows.** Which rows contain responses? Note that the task has more than one section, and trial numbering starts over in each.
- **Missing responses.** As with the stress data, decide how to handle timeouts, and make sure your filtering doesn't make that decision for you without your noticing.

Both data sets

- **Incomplete participants.** Did everyone finish both tasks? Decide on a rule for participants with missing data and apply it consistently across both data sets.
- **Participant IDs.** Do the same participants appear in both files, so that your final merge doesn't silently drop anyone?

Sanity check

The stress data have 38 participants with 80 trials each (48 distractor and 32 target items). After merging with the answers, you should have 3,040 response rows (or something near it; for instance, when I wrangled this data, I ended up with 3,033 rows), and no row should be missing an answer. If your numbers don't match (for instance, you end up with 2,000 rows), something went wrong upstream. Once you have the correct number of rows, you are ready to create a mean score.

Data Visualization

Make a plot of score taking into account item type (distractor vs target). Describe what you see. What is the apparent relationship between item type and score? Make a plot of the relationship

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between proficiency and score. Describe what you see. What is the apparent relationship between proficiency and score?

Data Analysis

Fit a linear model to mean score, including proficiency and type as predictors. Interpret the model output, to the best of your ability. What can we say about the relationship between our predictors and score? As part of your interpretation, explain what the intercept represents in this model. Is it a meaningful value, given the range of proficiency scores in your data? Finally, note that each participant contributes two mean scores to this model (one per item type). Briefly explain why that might be a problem for a linear model. We will return to this issue when we cover mixed-effects models.

AI Use

You may not upload this assignment and the associated data sets to an AI platform. You may prompt AI for help, with the following conditions: (1) beforehand, you should make an earnest effort to (re)solve data difficulties yourself (e.g., spend 10-15 minutes thinking through what might be necessary, exploring potential functions in R using ‘?’, reviewing class notes and Winter’s book); (2) you must transparently disclose AI use in your code.

Collaboration

You may collaborate with classmates so long as you work independently and come together to review and troubleshoot code. In other words, please do not subdivide this assignment in stages and copy code instead of making an attempt to author code yourself.

Submission

Upload a .Rmd (R markdown) file with complete, annotated code to your repo by the deadline. Your code should be well organized, reflecting the steps described in this assignment.